

P-values and Bayesian Analysis

Cristian Mesquida

Eindhoven University of Technology

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Two statistical philosophies

- ▶ Frequentist or Null hypothesis significance testing (NHST)
- ▶ Bayesian (Bayes theorem)

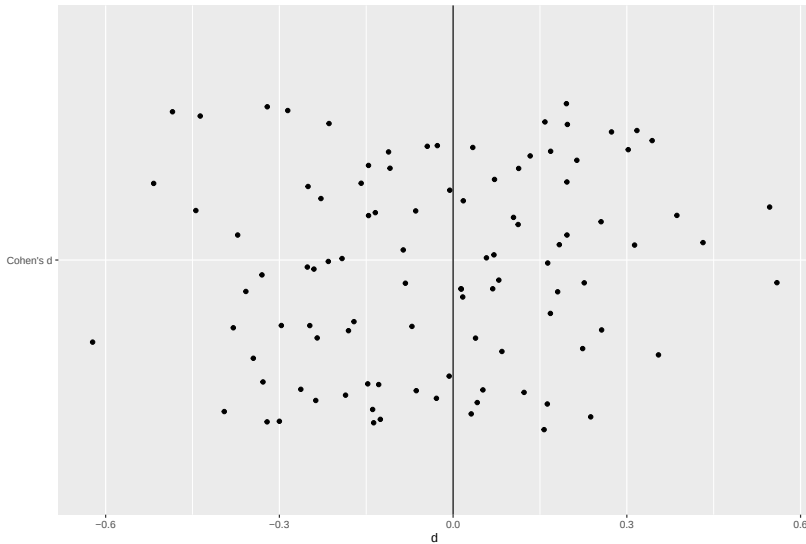
Frequentist

- ▶ Goal: not to make a fool out of yourself too often in the long run (error control)

Why to control for error rates?

- ▶ Any observed effect (difference between treatments, groups, or conditions) is influenced by sampling variation

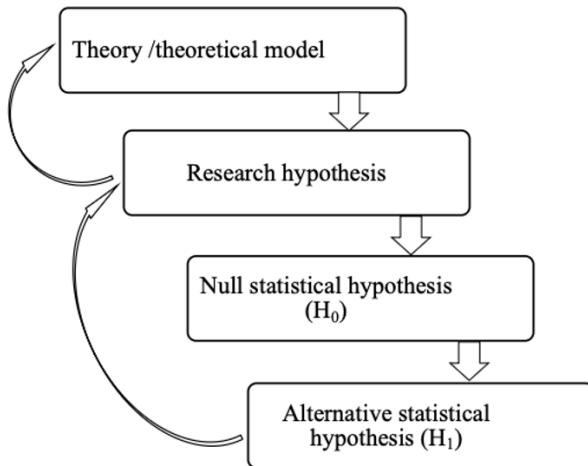
Distribution of effect sizes (two-sided t-test, $N = 60$ and a true effect of 0)



Why to control for error rates?

- ▶ We need a methodological procedure that allows us to make claim while controlling for error rates
- ▶ We want to make claims (e.g., this intervention is superior to a control intervention)

Null hypothesis significance testing (NHST)



4 states of the world

	Reject H0	Do not reject H0
H0 true (no effect)	Type I error (α)	Correct ($1 - \alpha$)
H1 true (effect)	Correct ($1 - \beta$)	Type II error (β)

α = probability of making a type I error β = probability of making a type II error power = $1 - \beta$

What will your next study show?

```
H0 <- 0.5 # probability of H0 being true  
H1 <- 0.5 # probability of H1 being true  
alpha <- 0.05 # type I error  
power <- 0.8 # power (1 - type II error)
```

	Reject H0	Do not reject H0
H0 true (no effect)	$0.5 * 0.05 = 0.025$	$0.5 * (1 - 0.05) = 0.475$
H1 true (effect)	$0.5 * 0.8 = 0.4$	$(1 - 0.8) * 0.5 = 0.1$

How to improve? Decrease type II error rate (increase power)

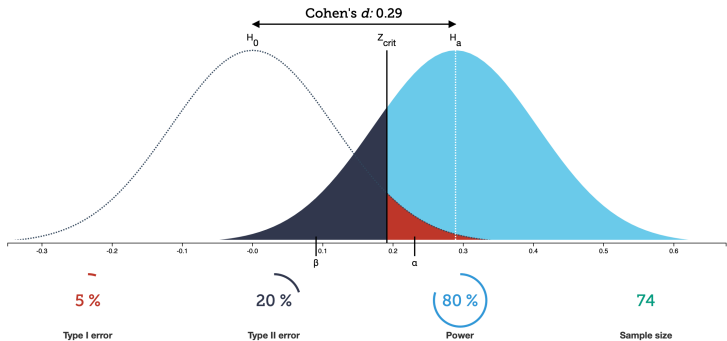
```
H0 <- 0.5 # probability of H0 being true
H1 <- 0.5 # probability of H1 being true
alpha <- 0.05 # type I error
power <- 0.9 # power (1 - type II error)
```

	Reject H0	Do not reject H0
H0 true (no effect)	$0.5 * 0.05 = 0.025$	$0.5 * (1 - 0.05) = 0.475$
H1 true (effect)	$0.5 * 0.9 = 0.45$	$(1 - 0.9) * 0.5 = 0.05$

How to improve? Decrease type I error rate

```
H0 <- 0.5 # probability of H0 being true
H1 <- 0.5 # probability of H1 being true
alpha <- 0.01 # type I error
power <- 0.9 # power ( 1 - type II error)
```

	Reject H0	Do not reject H0
H_0 true (no effect)	$0.5 * 0.01 = 0.005$	$0.5 * (1 - 0.01) = 0.495$
H_1 true (effect)	$0.5 * 0.9 = 0.45$	$(1 - 0.9) * 0.5 = 0.05$



From Kristoffer Magnusson's blog. Check it out to learn more about NHST.

Which error should the investigator aim for?

- ▶ It depends (Is it more serious to convict an innocent man, or to acquit a guilty?)
- ▶ What is most costly?
- ▶ Determining how the balance must be struck should be left to the investigator
- ▶ However, most investigators use an α level of 0.05 and power of 80% by default

Frequentist testing (example)

- ▶ H : there is a difference between the two groups (non-directional prediction)
- ▶ Set H_0 : $\mu_A - \mu_B = 0$ (but it could be any number, or a range and it should be the opposite as to your H)
- ▶ Set H_1 : $\mu_A - \mu_B \neq 0$ (non-directional buth it could be directional: $>$ (greater) or $<$ (smaller), depending on your H)
- ▶ Use a statistical test to calculate a test statistic
- ▶ Calculate a p -value: the probability of seeing data/test statistics as extreme or more extreme than the observed data with repeated sampling when H_0 is true

P -value as a rule to govern our behavior in the long run

- ▶ $p \leq \alpha$: Reject H_0
- ▶ $p > \alpha$: Fail to reject H_0
- ▶ When you act as if there is an effect when $p \leq 0.05$, in the long run you won't be wrong more than 5% of the time

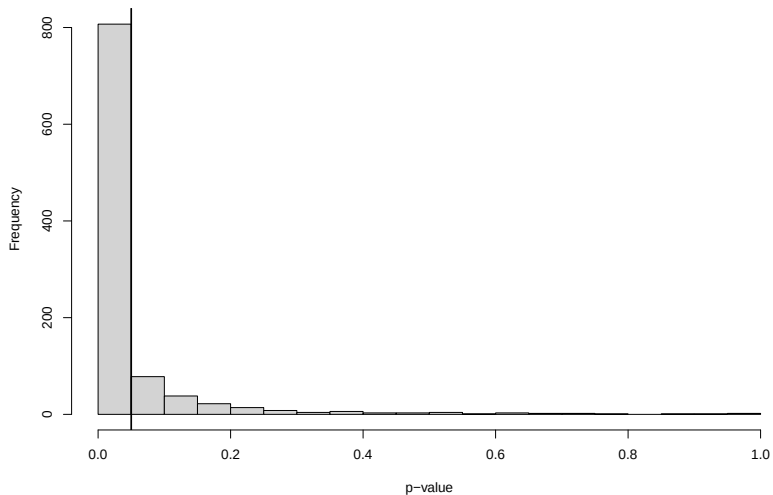
Interpretation of a p -value

- ▶ A p -value is the probability of getting the observed or more extreme data, assuming H_0 is true. In other words: $P(\text{data is extreme or more extreme} \mid H_0)$
- ▶ If you want to know the probability of H , we need Bayesian statistics
- ▶ Important! When $p > \alpha$, we can't conclude in favour of H_0 . We are looking for enough evidence to reject H_0 . If we fail to reject H_0 , we do not conclude H_0 is true. To accept H_0 , we need an equivalence test (next lecture)

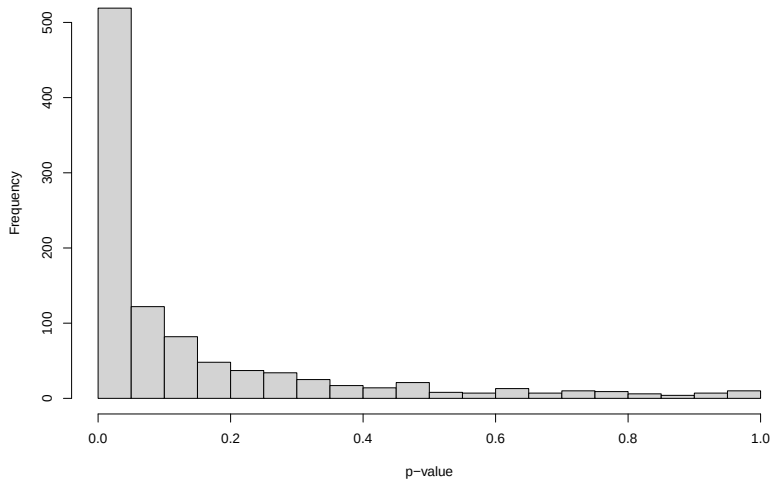
Frequentist interpretation of a p -value

- ▶ Error rates are frequentist concepts. They are about the long run!
- ▶ If I were to repeat the same experiment 1000 times, how many replications would yield a significant p -value?
- ▶ Check Magnusson's app out to see how statistical power affects the distribution of p -values

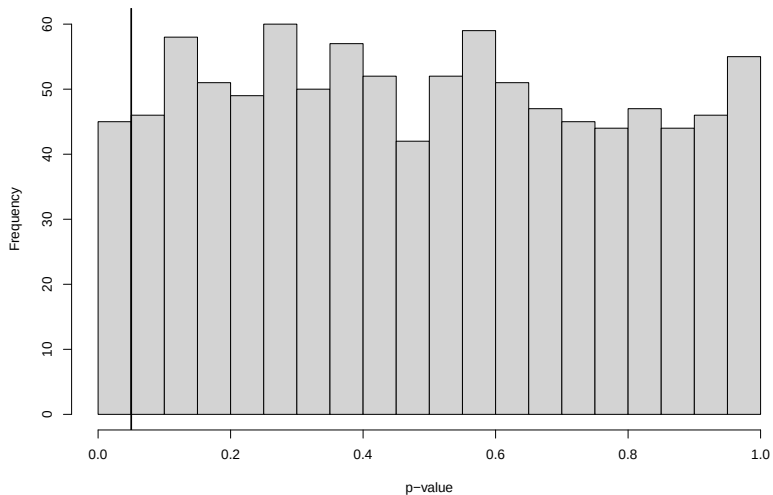
Distribution of p -values when power is 80%



Distribution of p -values power is 50%



Distribution of p -values when the null hypothesis is true



- ▶ Assuming H_0 is false, the higher the statistical power, the more left skewed the distribution of p -values is
- ▶ When H_0 is true, the power of the test corresponds to the type I error rate
- ▶ That is, 5% of p -values will be smaller than the type I error rate ($\alpha = 0.05$)

“no isolated experiment, however significant in itself, can suffice for the experimental demonstration of any natural phenomenon”

Fisher, 1937, p. 13

Summary of NHST

- ▶ Decide upon type I and type II error rates
- ▶ P -value is the probability of observing data (or more extreme) assuming H_0 is true
- ▶ A statistical test is used to compute a p -value
- ▶ Based on the p -value, decide to reject or fail to reject H_0
- ▶ NHST results in a binary decision (reject/not reject)
- ▶ Use NHST if you want to control how often you will be wrong in the long term

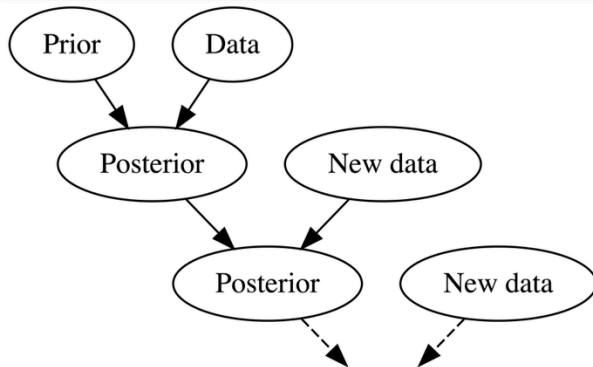
Bayesian analysis

- ▶ Bayesian statistics allows you to express evidence in terms of 'degrees of belief.'

Interpreting probability

- ▶ In the Bayesian philosophy, a probability measures the relative plausibility of an event
- ▶ In the frequentist philosophy, a probability is interpreted as the long-run relative frequency of a repeatable event.

Bayesian workflow



Bayesian workflow

A *prior* represents the information about an uncertain parameter that is combined with the probability distribution of **new data** to yield the **posterior** distribution

Bayes Rule

$$P(B|A) = \frac{P(B) * L(B|A)}{P(A)}$$

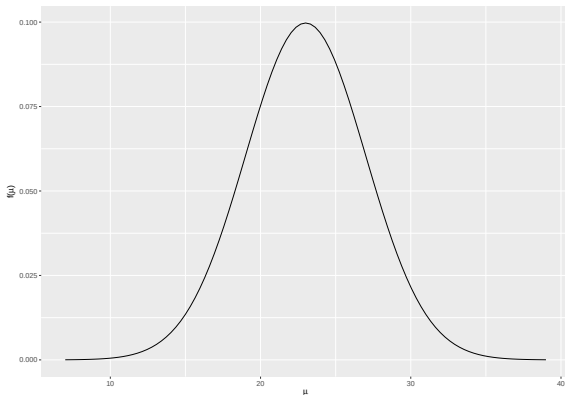
$$\text{posterior} = \frac{\text{priori} * \text{likelihood}}{\text{normalizing constant}}$$

The Prior

- ▶ The prior represents our understanding and uncertainty about an unknown parameter
- ▶ It is quantified using a prior probability distribution or range of values that the researcher considers credible before collecting data
- ▶ It is informed by previous studies or expert opinion: “today’s posterior is tomorrow’s prior” (Dennis Lindley)
- ▶ For a continuous outcome, we can model the prior as normal distribution. There are only two parameters to set: mean and standard deviation

Example of prior

- ▶ Suppose we want to know how much milk a cow produces on average per day
- ▶ We have 10 cows with an average production of 23L a day and a standard deviation of 4



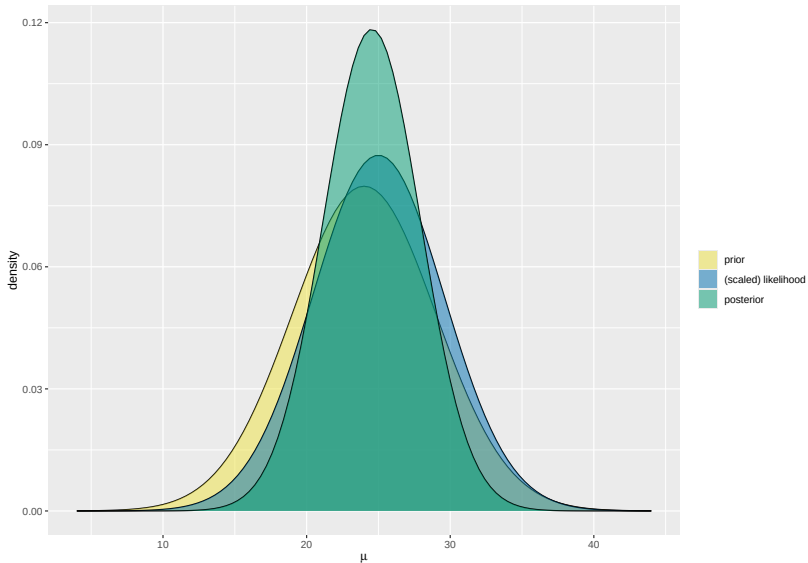
Prior, data and posterior

- ▶ Now suppose that we collect data from more farms
- ▶ We can use the Bayes theorem to estimate the posterior distribution

```
m1 <- 24          # prior mean
sd1 <- 5          # prior standard deviation
variance <- 25    # variance
m2 <- 25          # observed mean of produced milk at our farm
sample <- 30      # sample size of cows

plot <- plot_normal_normal(mean = m1, sd = sd1,
                           sigma = variance,
                           y_bar = m2, n = sample)
```

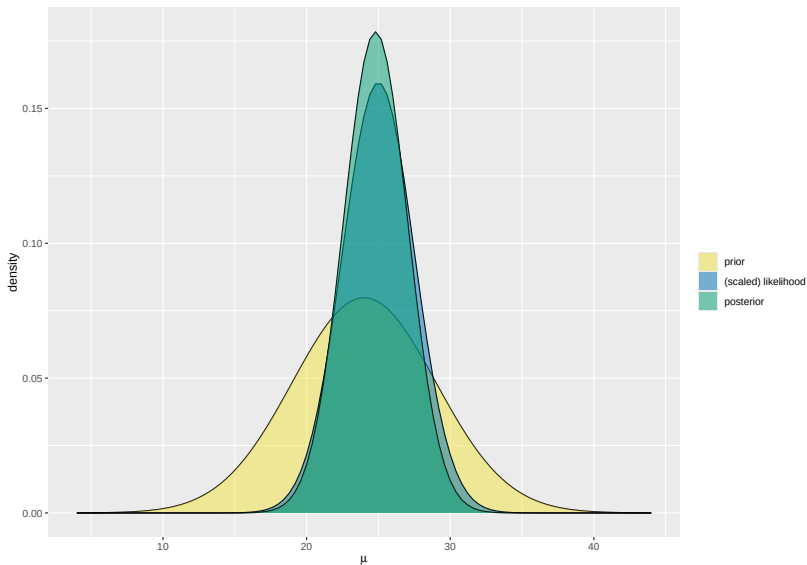
Plot the prior, data and posterior



Summary statistics

	model	mean	mode	var	sd
1	prior	24.00000	24.00000	25.00000	5.000000
2	posterior	24.54545	24.54545	11.36364	3.370999

What happens if we increase the sample size of cows?



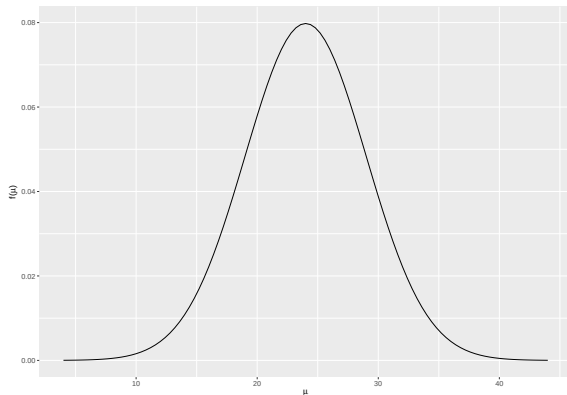
- ▶ As we collect more data, the posterior is dominated by data

Weak prior

- ▶ A weak prior is intentionally uninformative
- ▶ Still enforces basic constraints (e.g., physical limits)
- ▶ Has large variance $\rightarrow$ spreads values over a wide range
- ▶ Lets the data (likelihood) to dominate the posterior

Example of weak prior

- ▶ Milk production ≥ 0
- ▶ You're basically saying: "I don't know much, but negative milk output is impossible."

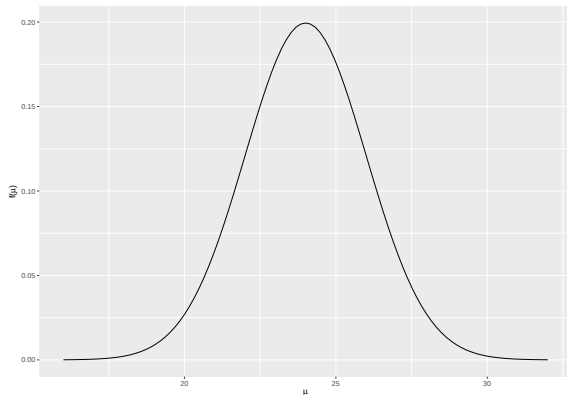


Strong prior

- ▶ A strong prior is highly informative
- ▶ Has small variance $\rightarrow$ tightly concentrated around certain values
- ▶ Encodes substantial prior knowledge or belief
- ▶ Can strongly influence the posterior (sometimes even dominate the data if data is limited)

Example of strong prior

- ▶ A dairy cow typically produces an average of nearly 8,900 liters of milk a year, or 24 L/day (source)
- ▶ You might set a prior tightly centered around 24 with low variance



Bayesian hypothesis testing

- ▶ Like in NHST, we compare two statistical hypotheses: H_0 and H_1
- ▶ Instead of using p -values, we use Bayes factors
- ▶ A Bayes factor (BF_{10}) compares the evidence provided by data for H_1 over H_0 :

$$BF_{10} = \frac{P(D|H_1)}{P(D|H_0)}$$

- ▶ A $BF_{10} = 5$ -> data are 5x more likely under H_1 than H_0
- ▶ So instead of “rejecting” H_0 , you say: “The data support H_1 over H_0 by this factor”
- ▶ A $BF_{10} = 0.1$ is evidence in favour of H_0 because $1/0.1 = 10$

- ▶ We can also compare H_0 against H_1
- ▶ BF_{01} compares the evidence provided by data for H_0 over H_1 :



$$BF_{01} = \frac{P(D|H_0)}{P(D|H_1)}$$

- ▶ A $BF_{01} = 5 \rightarrow$ data are 5x more likely under H_0 than H_1
- ▶ Check Magnusson's app out to gain intuition about Bayesian hypothesis testing

Example of Bayesian two-sided t -test

- ▶ Suppose we expect that milk productivity will vary between two cow breeds (Friesian vs. Hereford), where:
- ▶ $H_0: \mu_F - \mu_H = 0$
- ▶ $H_1: \mu_F - \mu_H \neq 0$
- ▶ The prior for H_0 is a point-null hypothesis assuming the difference is 0
- ▶ The prior for the alternative is modelled as a Cauchy distribution
- ▶ The calculated BF tells you how much you should shift your belief towards one hypothesis compared to another
- ▶ In the Bayesian t -test function, the main new argument is `rscale` which sets the width of the prior distribution around the alternative hypothesis.

Conduct a Bayesian two-sided t -test: simulate data

```
set.seed(124) # for reproducibility

# Load packages
library(BayesFactor)
library(bayestestR)

# Simulate data for a two independent groups
group <- rep(c("friesian", "hereford"), each = 50)

data <- data.frame(
  group = group,
  milk = ifelse(group == "friesian",
                rnorm(100, mean = 28, sd = 3),
                rnorm(100, mean = 26, sd = 3))
)
```

Bayesian two-sided *t*-test

```
ttestBF(formula = milk ~ group,  
        data = data,  
        rscale = "medium", # prior distribution around H1  
        paired = FALSE)
```

Bayes factor analysis

[1] Alt., $r=0.707$: 1030.96 $\pm 0\%$

Against denominator:

Null, $\mu_1 - \mu_2 = 0$

Bayes factor type: BFindepSample, JZS

Description of the output

- ▶ Note that “Alt.” denotes H_1 so it's measuring BF_{10}
- ▶ With the “medium” prior, we have a BF_{10} of 1031, suggesting H_1 is 1031 times more likely than H_0
- ▶ By the guidelines from Wagenmakers et al. (2011), this is very strong evidence
- ▶ The percentage (%) next to the BF is the proportional error estimate and tells you the error in estimating the Bayes factor value
- ▶ Less error is better and a rough rule of thumb is less than 20% is acceptable (van Doorn et al. 2021)
- ▶ In this example, the error estimate is 0%. This means we precisely estimate BF

Example of Bayesian one-sided t -test

- ▶ Suppose we expect Friesian cows are more productive than Aberdeen:
- ▶ $H_0: \mu_F - \mu_A \leq 0$
- ▶ $H_1: \mu_F - \mu_A > 0$

Conduct a Bayesian one-sided t -test: simulate data

```
set.seed(123) # for reproducibility

# Simulate data
group <- rep(c("friesian", "aberdeen"), each = 50)

data <- data.frame(
  group = group,
  milk = ifelse(
    group == "friesian",
    rnorm(n = 100, mean = 28, sd = 3),
    rnorm(n = 100, mean = 22, sd = 3))
)
```

Bayesian one-tailed *t*-test

```
ttestBF(formula = milk ~ group,  
        data = data,  
        rscale = "medium",  
        paired = FALSE,  
# only positive as we expect Friesian > Aberdeen  
        nullInterval = c(0, Inf))
```

Bayes factor analysis

```
-----  
[1] Alt., r=0.707 0<d<Inf      : 6063.901      ±0%  
[2] Alt., r=0.707 !(0<d<Inf) : 2.490859e+15 ±0%
```

Against denominator:

Null, $\mu_1 - \mu_2 = 0$

Bayes factor type: BFindepSample, JZS

Summary of Bayesian statistics

- ▶ Probabilistic framework for learning from data
- ▶ Based on Bayes' theorem
- ▶ Updates prior beliefs into posterior distributions using observed data
- ▶ The influence of prior vs data depends on:
- ▶ With weak priors + large data $\rightarrow$ data (likelihood) dominates the posterior
- ▶ With informative priors + small data $\rightarrow$ prior dominates the posterior
- ▶ Treats parameters as uncertain quantities (distributions) rather than fixed values
- ▶ Uses Bayes Factor (BF_{10}) to quantify evidence of H_1 over H_0
- ▶ Allows evidence instead of binary decisions (reject/not reject)

Additional resources

- ▶ Improving your statistical inferences; Chapters 1, 2 3 and 4
- ▶ PsychR
- ▶ If you are keen to learn more about Bayesian analysis, see the Bayes Rules! book

References

- van Doorn, Johnny, Don van den Bergh, Udo Böhm, Fabian Dablander, Koen Derks, Tim Draws, Alexander Etz, et al. 2021. "The JASP Guidelines for Conducting and Reporting a Bayesian Analysis." *Psychonomic Bulletin & Review* 28 (3): 813–26. <https://doi.org/10.3758/s13423-020-01798-5>.
- Wagenmakers, Eric-Jan, Ruud Wetzels, Denny Borsboom, and Han L. J. van der Maas. 2011. "Why Psychologists Must Change the Way They Analyze Their Data: The Case of Psi: Comment on Bem (2011)." *Journal of Personality and Social Psychology* 100 (3): 426–32. <https://doi.org/10.1037/a0022790>.